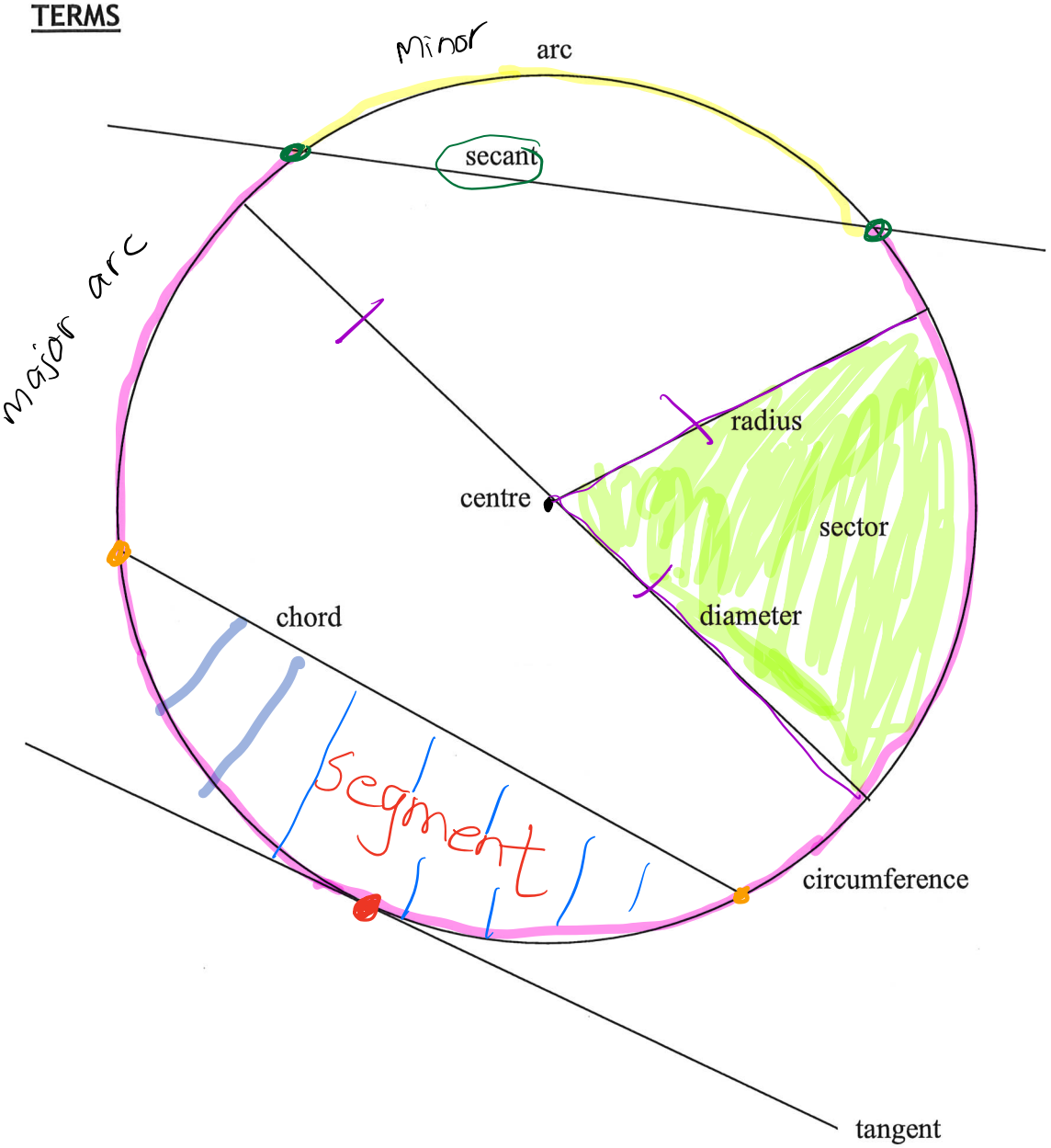
CIRCLE GEOMETRY

TERMS



It is always useful to trace the origins of words, even in Maths! In this way words that seem to be meaningless acquire a meaning.

arc: from the Latin *arcus* 'a bow'
centre: from the Latin *centrum* 'a sharp point'
chord: from the Greek *chorde* 'a string' (as in a harp)
circumference: from the Latin *circum* 'around' and *ferre* 'to carry'
diameter: from the Greek *dia* 'through, across' and *metron* 'a measure'
radius: from the Latin *radius* 'a rod, spoke, ray'
secant and **sector**: from the Latin *secare* 'to cut'
tangent: from the Latin *tangere* 'to touch'

GEOGEBRA INVESTIGATIONS

All the investigations below can be found at the following (case-sensitive!) web address:

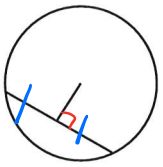
https://www.geogebra.org/m/dxun9cab

An Investigation via Geogebra

1) A circle has been drawn with a chord inside it and part of a radius going to the chord. **(GEOGEBRA INVESTIGATION 1)**

2) Try and position the radius so that it cuts the chord in half (meets at the midpoint of the chord).

4) Your figure should look something like this:



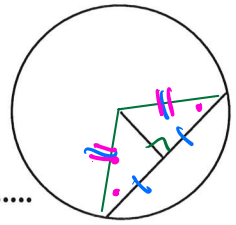
5) What is the size of the angle between the chord and the line coming from the centre when going to the midpoint of the chord? 90°

6) Use the geogebra page to experiment....
Will you get the same result with a circle of different size? Yes
A chord of diferent length? Yes

7) **What conclusion can you make based on this investigation?** If the line from the centre bisects the chord, and is perpendicular to the chord

8) Making a clever construction see if you can prove that your theorem will be true in every case. (Hint: make use of congruency.)

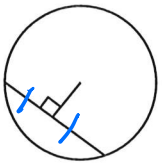
SAS, SSS, RHS, SAA



9) Now investigate further.....
If you start with the circle and the perpendicular to the chord.....

Will the perpendicular pass through the centre of the circle?

Yes



10) **What conclusion can you make based on this investigation?**

If a line from centre is perpendicular to a chord, it will bisect the chord

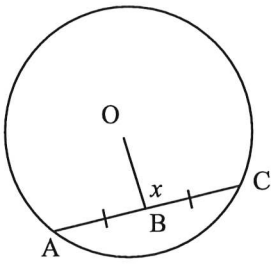
In both cases we land up working with a right angle, a radius and part of a chord.....i.e. 2 sides of a triangle. When we apply these concepts it be in using **Pythagoras.**

Exercise 2

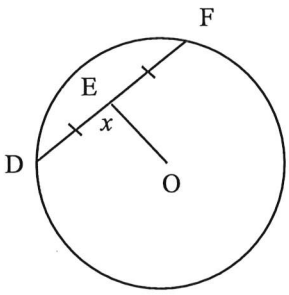
Giving reasons, find the values of the angles and/or sides marked with a letter of the alphabet in each of the following, assuming always that O is the centre of the circle.

1.

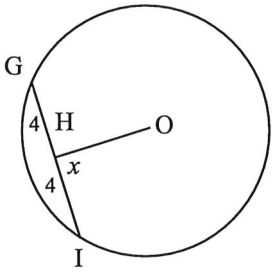
$x = 90°$ (line from centre to midpoint of chord)



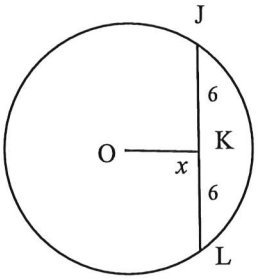
2.



3.

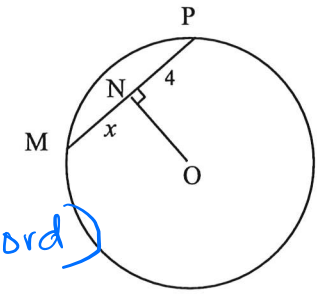


4.

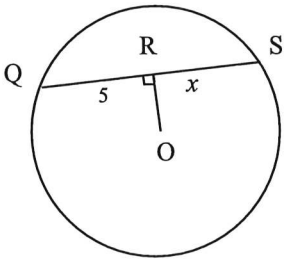


5.

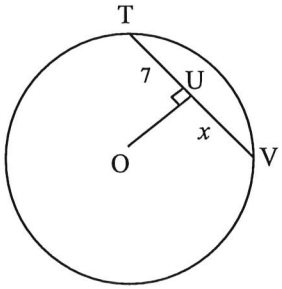
$x = 4$ units (line from centre ⊥ chord)



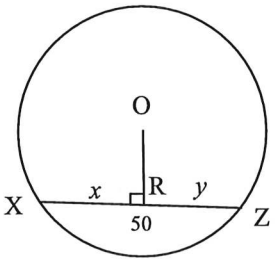
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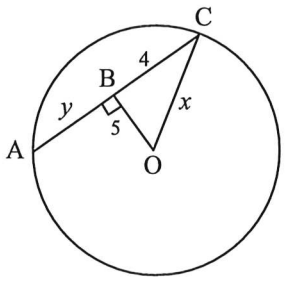
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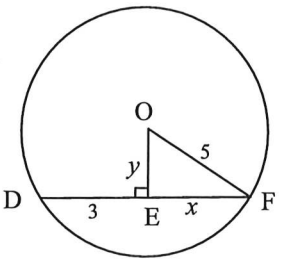
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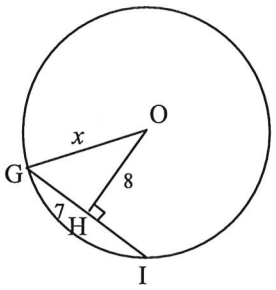
9.



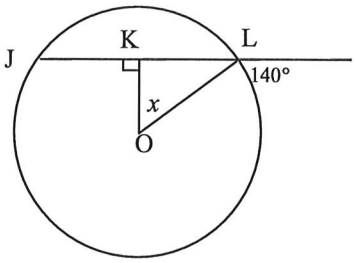
10.



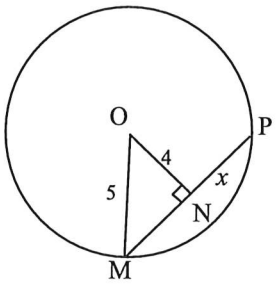
11.



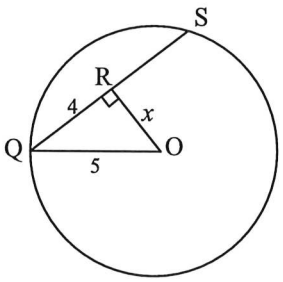
12.



13.



14.



An Investigation using Geogebra

Refer to the Geogebra **investigation 2.**

You are given a circle with centre marked C. Two points B and A are on the circumference of the circle. i.e. AC and BC are radii.

1. What is the size of $B\hat{C}A$ that you have chosen?

2. Move A and then B along the edge of the circle. What happens to $\hat{C}$ (the angle at the centre)?

3. Chord AB makes $\hat{D}$ anywhere in the circle. Move D around the circle and see what happens. Does the size of $\hat{C}$ change?

4. Move D to anywhere in the circle. What is the size of $\hat{D}$ that you have chosen?

5. Is there a relationship between the angle at the centre and $\hat{D}$?

6. Now move $\hat{D}$ so that it lies on the circumference. Does $\hat{D}$ get bigger or smaller as it moves towards the circumference?

7. What size is $\hat{D}$ now that you have placed D on the circumference?

8. Now calculate the value of the angle at the centre when D is on the circumference.

9. If D is moved along the circumference, does it change in size?

10. Do you notice anything about the size of $\hat{D}$ relative to the size of $\hat{C}$?

11. Move D off the circle (outside of the circle). What happens to $\hat{D}$ as you move it further away from the circumference?

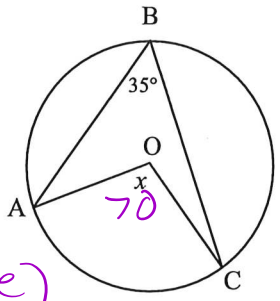
12. Describe the relationship between the angle at the centre and the angle at the circumference.

Exercise 3

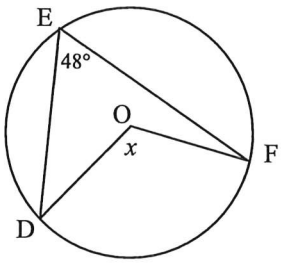
Giving reasons, find the values of the angles and/or sides marked with a letter of the alphabet in each of the following, assuming always that O is the centre of the circle.

1.

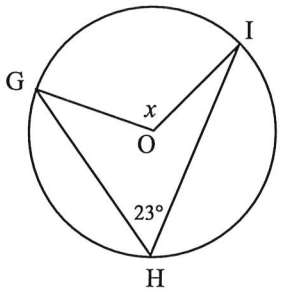
$x = 70°$
(∠ at centre = 2x ∠ at the circumference)



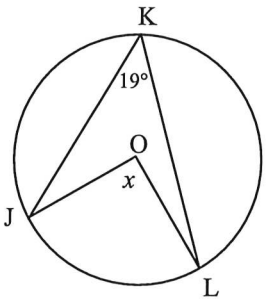
2.



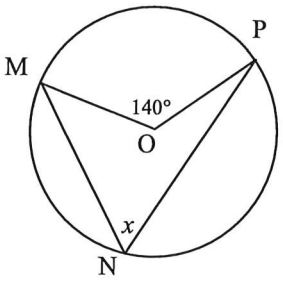
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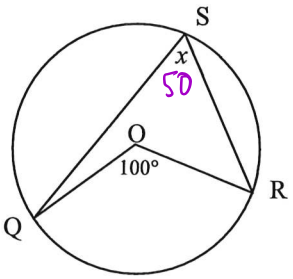
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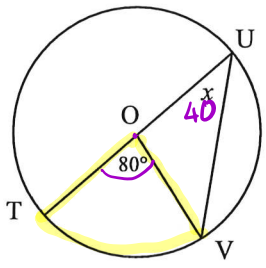
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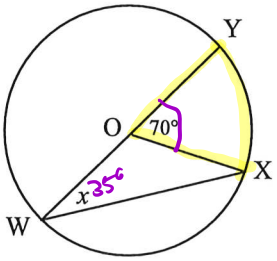
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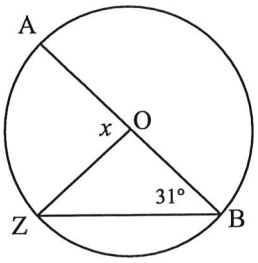
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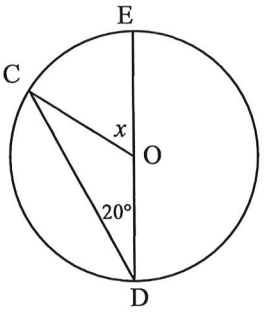
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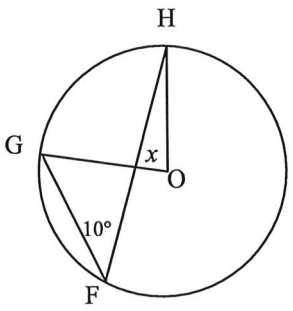
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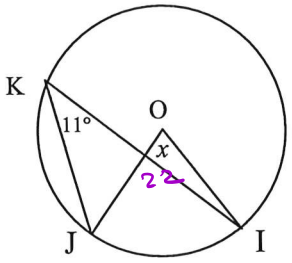
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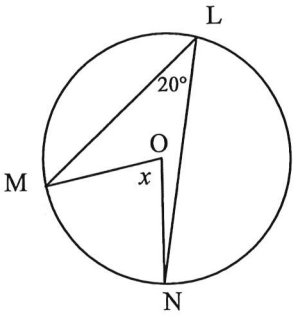
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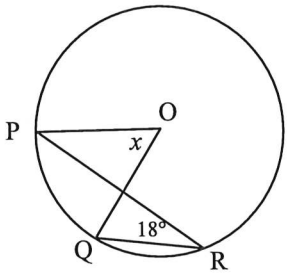
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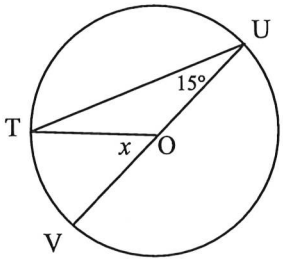
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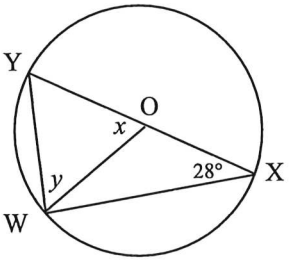
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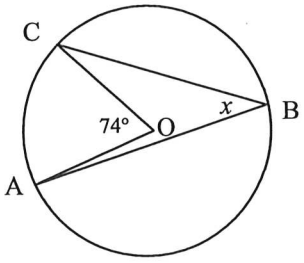
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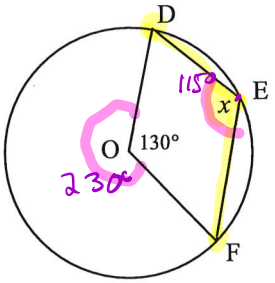
16.



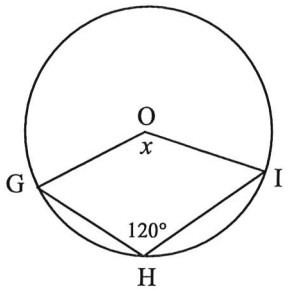
17.



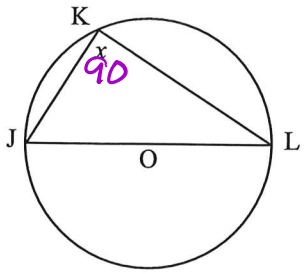
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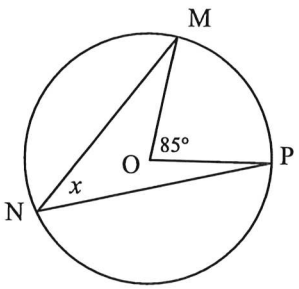
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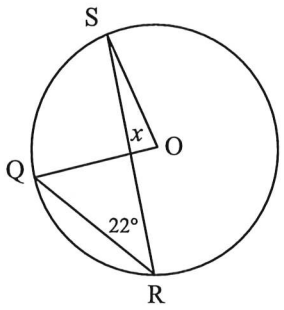
20. ∠ in semi circle



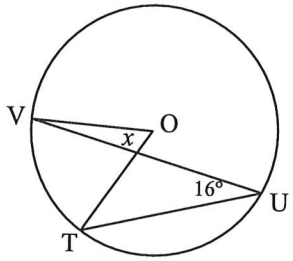
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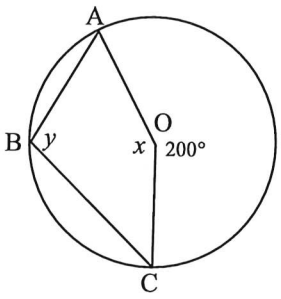
22.



23.

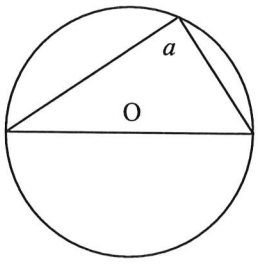


24.



An Investigation

1) Refer to the previous Geogebra investigation and **investigation 3.**
2) Make the angle at the centre measure ~~90°~~ 180°
3) How big is the angle at the circumference which is made by the same chord?

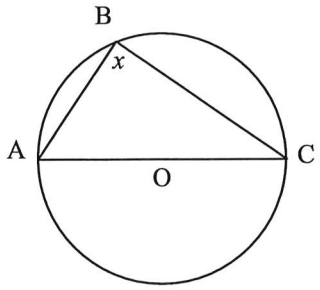


4) What can we conclude about the size of the angle made at the edge of the circle by any diameter?

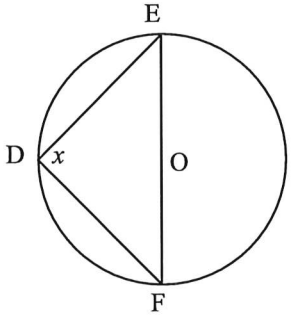
EXERCISE 4

Giving reasons, find the values of the angles and/or sides marked with a letter of the alphabet in each of the following, assuming always that O is the centre of the circle.

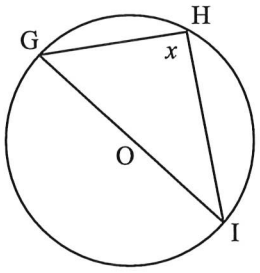
1.



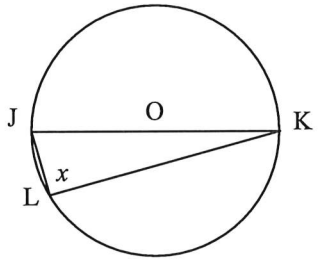
2.



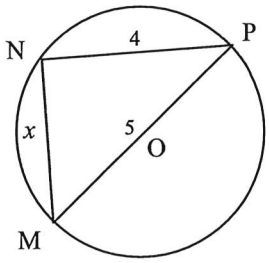
3.



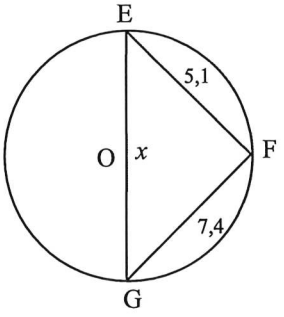
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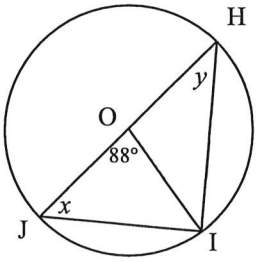
5.



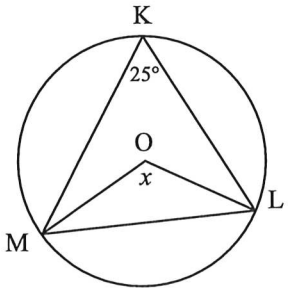
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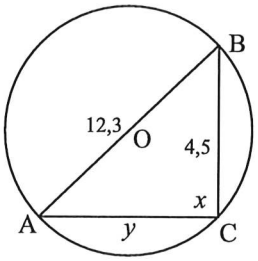
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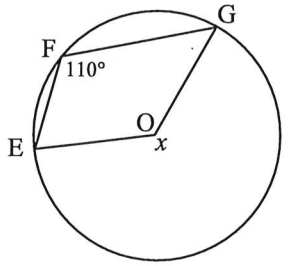
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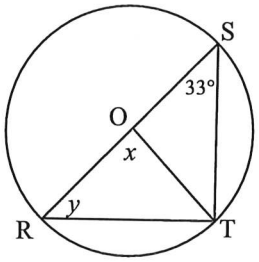
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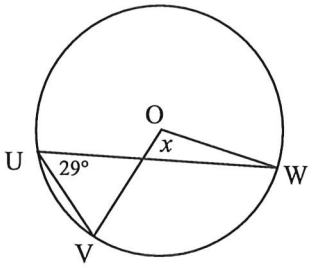
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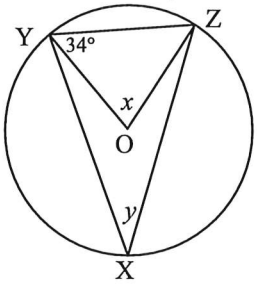
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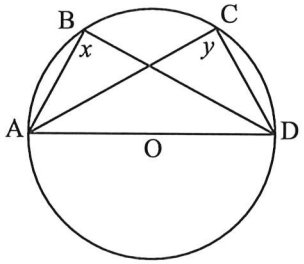
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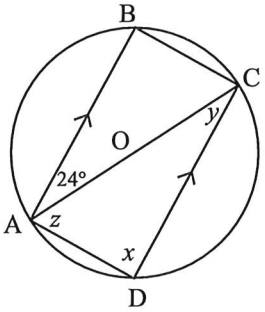
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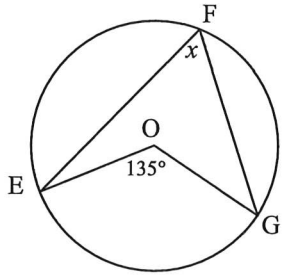
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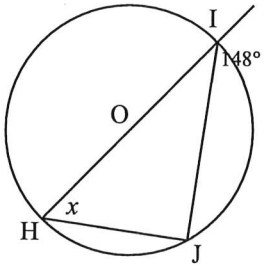
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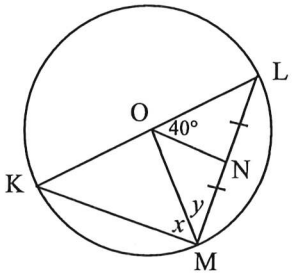
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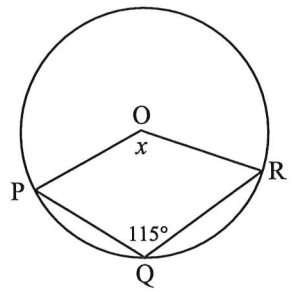
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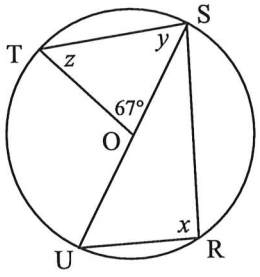
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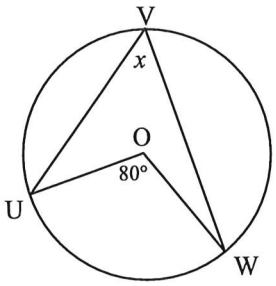
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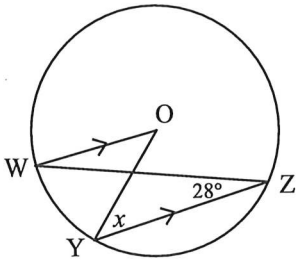
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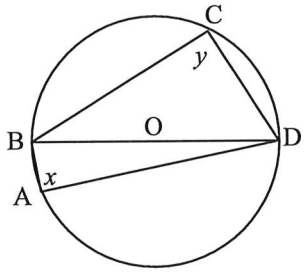
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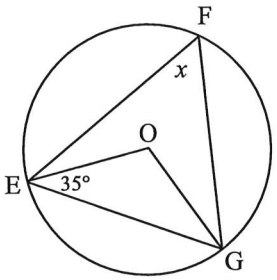
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23.



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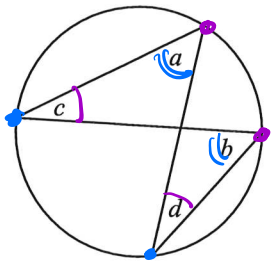


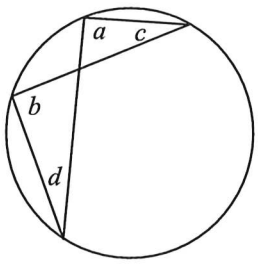
An Investigation

1) Similar diagrams have been drawn for you in Geogebra. Open the next Geogebra **Investigation 4. "Angles in the same segment".**

$a = b$
$c = d$

∠'s in same segment are equal





2) What do you notice about angles $\hat{C}$ and $\hat{E}$?

3) Drag point $\hat{C}$ along the circumference towards point $\hat{E}$. What do you notice about the size of $\hat{C}$ as you do so ?

4) Now drag $\hat{C}$ beyond $\hat{E}$ and towards $\hat{D}$. What do you notice about the size of $\hat{C}$ as you do so ?

5) Now drag $\hat{E}$ between $\hat{B}$ and $\hat{D}$. What do you notice?

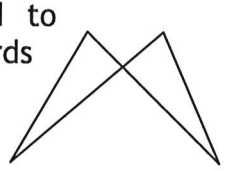
6) Open Geogebra **investigation 5.**

7) The point A is no longer on the circumference of the circle - What do you notice about the size of $\hat{A}$ and $\hat{E}$?

8) Now drag point A so that is lies outside the circle. - What do you notice about the size of $\hat{A}$ and $\hat{E}$?

9) Where must $\hat{A}$ and $\hat{E}$ lie so that they are the same size?

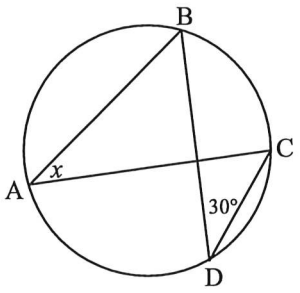
10) Consider whether you can use the information you have obtained to prove that a figure such as this one, lies on a circle (in other words that all four vertices lie on the circumference of a circle). What conditions must this figure fulfill before you can draw a circle around it?



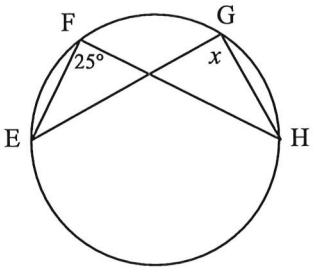
Exercise 5

Giving reasons, find the values of the angles and/or sides marked with a letter of the alphabet in each of the following, assuming always that O is the centre of the circle.

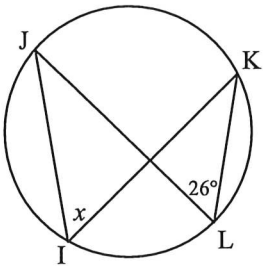
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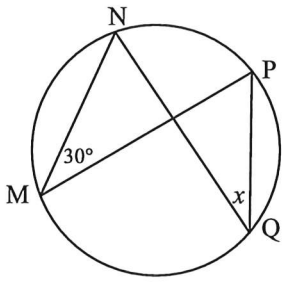
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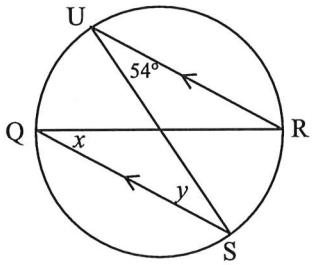
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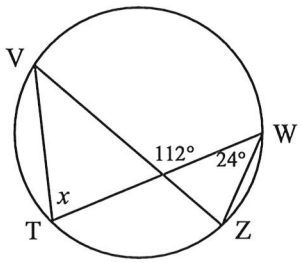
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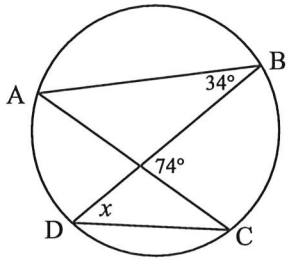
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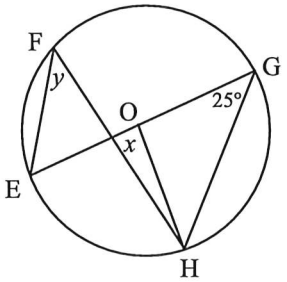
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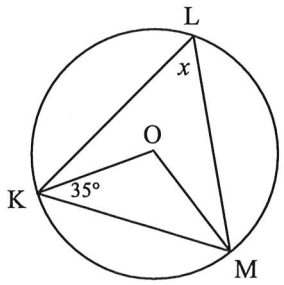
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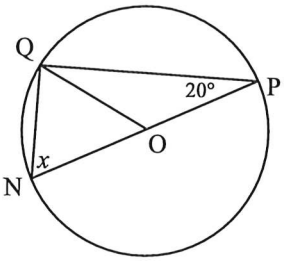
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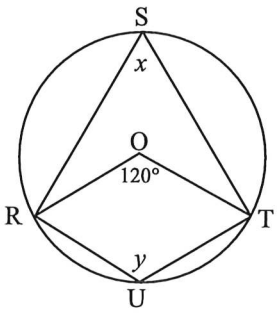
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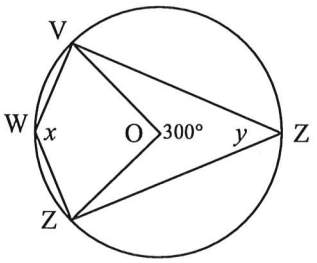
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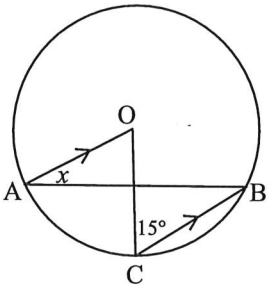
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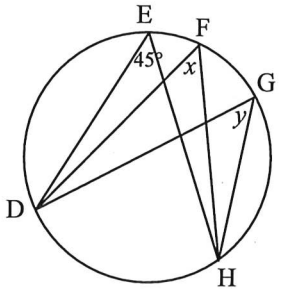
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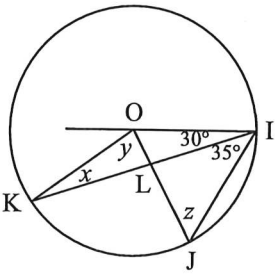
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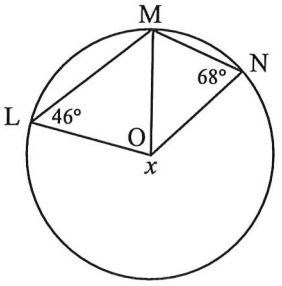
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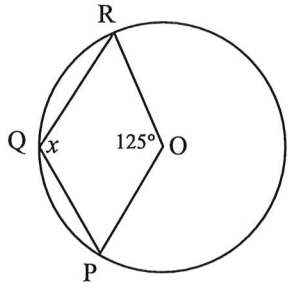
15.



16.



17.



18.

